

CHARAN YAKASIRI

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PROFESSIONAL SUMMARY

Building at the boundary of hardware and software, with hands-on experience in FPGA development, RTL, custom IP, and RISC-V. Deeply interested in low-level systems and understanding how digital systems work beneath the software layer.

PROFESSIONAL EXPERIENCE

Spanidea Systems

RTL FPGA Design Engineer Intern

Chennai, India

SEP 2026 - Present

- Designed a custom RTL controller for a vendor Avalon I2C master IP, integrated via Platform Designer and built through the Quartus flow; verified in QuestaSim (4 test cases passing) at **100 and 400 kHz**.
- Built a MicroBlaze system with UART16550 IP at **100 MHz** in Vivado / Vitis; simulated firmware via ELF files, then brought up the board and validated on hardware. Now extending the design to DDR via MIG IP.

SKILLS

Core Skills: RTL Design, FPGA Synthesis & Implementation, Timing Analysis, CDC, IP Integration, HW-SW Co-Design

Languages: Verilog HDL, SystemVerilog, C, Python, TCL

Technologies: AXI4-Lite/Stream, Avalon, I2C, UART, SPI, MicroBlaze, RISC-V

Tools: Vivado, Vitis, Intel Quartus, Platform Designer, QuestaSim, ModelSim, Openlane, Yosys, nextpnr

Boards: PYNQ-Z2, Arty-7, Spartan-7, VSDSquadron FM, Vicharak Shrike Lite

Other: Linux, Git, Shell Scripting

PROJECTS

FPGA-Based Multi-Camera Aggregator IP Core (PYNQ-Z2) | [GitHub](#)

- Architected a *dual-camera video aggregation pipeline* in PL fabric; designed 7 custom RTL IPs including a Sync Unit, Arbitration FSM, Frame Alignment Buffer, and AXI-Lite Register Bank.
- Managed CDC between a 74.25 MHz HDMI clock and 100 MHz AXI system clock using an asynchronous FIFO with Gray-code pointer synchronization (<0.1% frame loss); executed its complete [RTL-to-GDSII flow](#) using Cadence Genus and Innovus.
- Eliminated AXI crossbar address collisions by migrating to AXI SmartConnect for parallel HP port binding, enabling runtime frame-boundary-safe camera switching via PS over AXI-Lite MMIO.

RTL-Based CNN Accelerator on FPGA (PYNQ-Z2) | [GitHub](#)

- Led HW-SW co-integration and custom IP packaging for a CNN accelerator, leveraging AXI-Stream and AXI-Lite interfaces to achieve an optimized 0.89 ms system latency and 1121 FPS throughput.
- Developed Python-based image preprocessing and postprocessing pipelines, establishing an automated validation framework to verify PS-PL data transfer integrity over 1000+ executions.

RISQ: FPGA-Based, RISC-V-Triggered Qubit State Discriminator | In Progress

- Architecting a RISC-V-triggered FPGA system for real-time qubit state discrimination on a PYNQ-Z2-class board, in a two-person team; own the RISC-V control core, the decision unit, and the on-chip validation logic.
- Designing a custom hardware decision path tightly coupled to the processor pipeline for low-latency readout classification, with accuracy and latency measured on-chip. Architecture finalized; RTL implementation starting.

EDUCATION

B.E – Electronics and Communication Engineering

Saveetha Engineering College, Chennai

JUN 2023 - MAY 2027

GPA: 8.0

Higher Secondary Education – PCM (Class XII)

Terapanth Jain Vidyalaya Mat. Hr. Sec. School, Chennai

JUN 2021 - MAR 2023

GPA: 8.5

CERTIFICATIONS & ACHIEVEMENTS

- DVCON (4th Round)** – Reached Round 4 of the international design competition with a RISC-V + FPGA task-aware object detection architecture
- Samsung ISWDP Fellow (Cohort 4)** – Gained exposure to semiconductor manufacturing and Synopsys TCAD tools
- FPGA Design Program – VSD** – Built and integrated custom IPs (GPIO, PWM) with RISC-V on VSDSquadron FM
- RTOS using QNX – Pi Square Technologies** – Studied QNX microkernel architecture using Momentics IDE